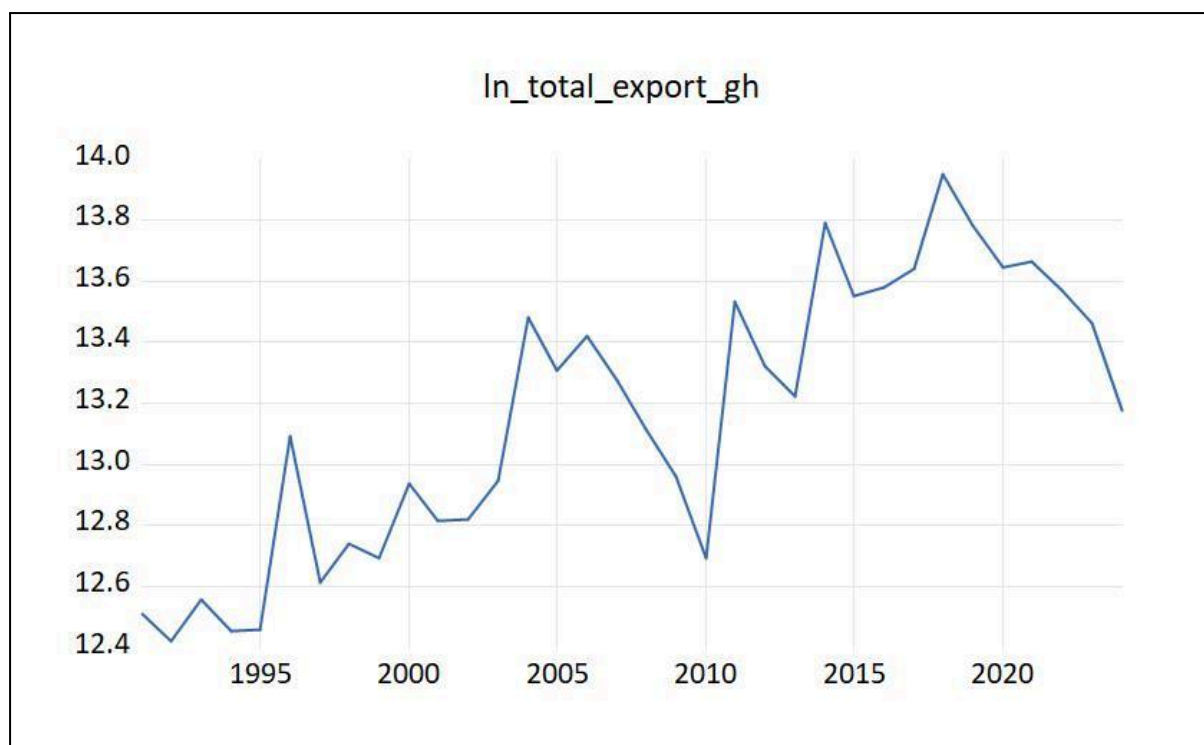
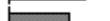
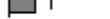
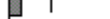




Date: 09/26/26 Time: 19:51

Sample: 1991 2024

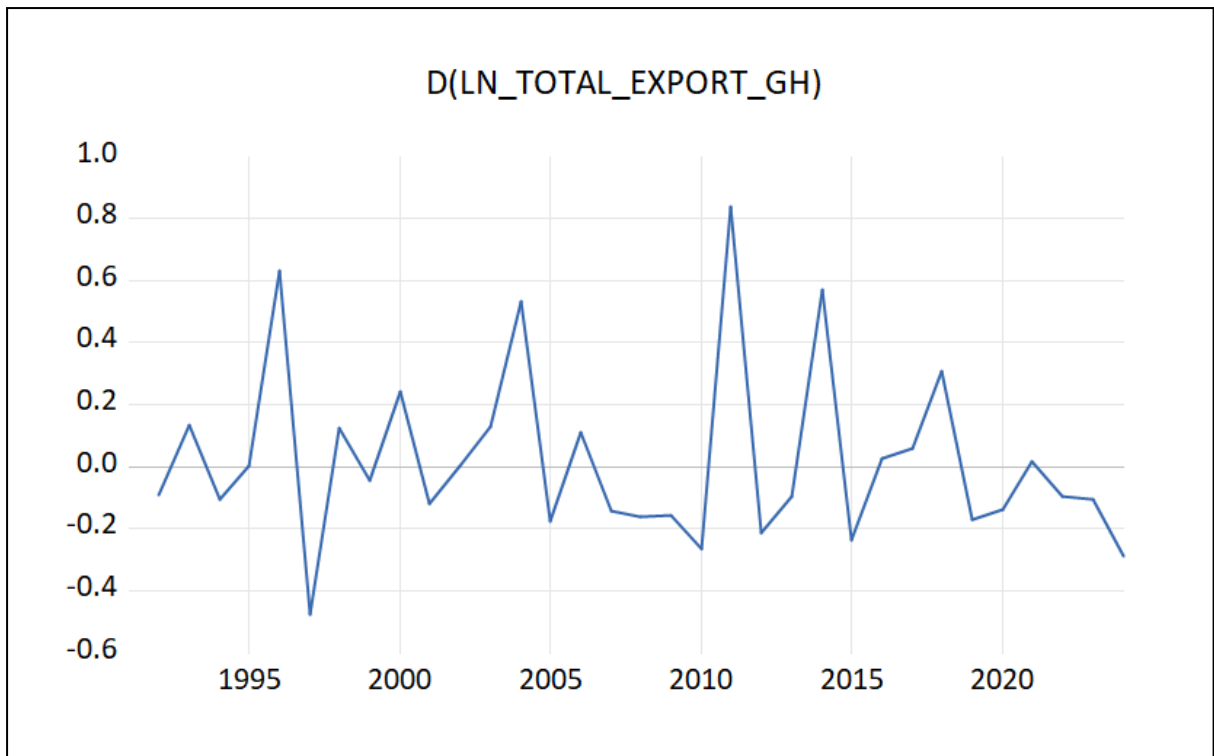
Included observations: 34

Included observations: 64					AC	PAC	Q-Stat	Prob
Autocorrelation	Partial Correlation							
		1	0.773	0.773	22.181	0.000		
		2	0.679	0.201	39.814	0.000		
		3	0.610	0.090	54.500	0.000		
		4	0.477	-0.148	63.801	0.000		
		5	0.357	-0.120	69.166	0.000		
		6	0.310	0.082	73.373	0.000		
		7	0.298	0.163	77.395	0.000		
		8	0.220	-0.087	79.668	0.000		
		9	0.101	-0.282	80.168	0.000		
		10	0.080	0.022	80.495	0.000		
		11	0.025	0.055	80.528	0.000		
		12	-0.058	-0.018	80.713	0.000		
		13	-0.103	-0.106	81.330	0.000		
		14	-0.079	0.053	81.715	0.000		
		15	-0.048	0.192	81.866	0.000		
		16	-0.095	-0.079	82.481	0.000		

Lo primero que hago son los test de Dickey Fuller para comprobar la existencia de raíz unitaria:

Null Hypothesis: LN TOTAL EXPORT GH has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-3.445559	0.0625
Test critical values:	1% level		-4.262735	
	5% level		-3.552973	
	10% level		-3.209642	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN TOTAL EXPORT GH)				
Method: Least Squares				
Date: 09/26/26 Time: 19:47				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN TOTAL EXPORT GH(-1)	-0.673243	0.195394	-3.445559	0.0017
C	8.462324	2.434370	3.476186	0.0016
@TREND("1991")	0.024270	0.009253	2.622779	0.0136
R-squared	0.293695	Mean dependent var		0.020227
Adjusted R-squared	0.246608	S.D. dependent var		0.285825
S.E. of regression	0.248091	Akaike info criterion		0.136462
Sum squared resid	1.846468	Schwarz criterion		0.272508
Log likelihood	0.748375	Hannan-Quinn criter.		0.182238
F-statistic	6.237285	Durbin-Watson stat		1.901546
Prob(F-statistic)	0.005431			

Como el coeficiente de tendencia es significativo comparo el estadístico t con el estadístico tau al 5%, como $-3.44 > -3.55$ NO rechazo la hipótesis nula de raíz unitaria, por lo que diferencio la serie una vez:



Compruebo nuevamente los test de Dickey Fuller:

Null Hypothesis: D(LN_TOTAL_EXPORT_GH) has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-8.136367	0.0000
Test critical values:	1% level		-4.273277	
	5% level		-3.557759	
	10% level		-3.212361	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN_TOTAL_EXPORT_GH,2)				
Method: Least Squares				
Date: 10/01/26 Time: 19:44				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_TOTAL_EXPORT_GH(-1))	-1.398761	0.171915	-8.136367	0.0000
C	0.128604	0.103699	1.240171	0.2249
@TREND("1991")	-0.005319	0.005222	-1.018600	0.3168
R-squared	0.696124	Mean dependent var		-0.006200
Adjusted R-squared	0.675167	S.D. dependent var		0.477499
S.E. of regression	0.272146	Akaike info criterion		0.324105
Sum squared resid	2.147842	Schwarz criterion		0.461517
Log likelihood	-2.185674	Hannan-Quinn criter.		0.369653
F-statistic	33.21684	Durbin-Watson stat		2.162288
Prob(F-statistic)	0.000000			

Como el coeficiente de tendencia no es significativo pruebo el test sin tendencia y con intercepto:

Null Hypothesis: D(LN TOTAL EXPORT GH) has a unit root
 Exogenous: Constant
 Lag Length: 0 (Automatic - based on SIC, maxlag=8)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-8.081732	0.0000
Test critical values:		
1% level	-3.653730	
5% level	-2.957110	
10% level	-2.617434	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
 Dependent Variable: D(LN TOTAL EXPORT GH,2)
 Method: Least Squares
 Date: 09/26/26 Time: 19:59
 Sample (adjusted): 1993 2024
 Included observations: 32 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN TOTAL EXPORT GH(-1))	-1.387209	0.171648	-8.081732	0.0000
C	0.035177	0.048411	0.726638	0.4731
R-squared	0.685252	Mean dependent var		-0.006200
Adjusted R-squared	0.674761	S.D. dependent var		0.477499
S.E. of regression	0.272316	Akaike info criterion		0.296757
Sum squared resid	2.224686	Schwarz criterion		0.388365
Log likelihood	-2.748110	Hannan-Quinn criter.		0.327123
F-statistic	65.31440	Durbin-Watson stat		2.105986
Prob(F-statistic)	0.000000			

El intercepto tampoco es significativo, por lo que hago el test sin intercepto ni tendencia:

Null Hypothesis: D(LN_TOTAL_EXPORT_GH) has a unit root				
Exogenous: None				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-8.112022	0.0000
Test critical values:	1% level		-2.639210	
	5% level		-1.951687	
	10% level		-1.610579	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN_TOTAL_EXPORT_GH,2)				
Method: Least Squares				
Date: 09/27/26 Time: 15:16				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_TOTAL_EXPORT_GH(-1))	-1.374019	0.169381	-8.112022	0.0000
R-squared	0.679713	Mean dependent var		-0.006200
Adjusted R-squared	0.679713	S.D. dependent var		0.477499
S.E. of regression	0.270235	Akaike info criterion		0.251704
Sum squared resid	2.263841	Schwarz criterion		0.297508
Log likelihood	-3.027263	Hannan-Quinn criter.		0.266887
Durbin-Watson stat	2.089343			

Aca directamente comparo el t estadístico con el estadístico tau al 5% o simplemente observo el p-valor asociado al test y veo que rechazo la hipótesis nula incluso al 1% de significatividad, por lo que la serie diferenciada no tiene raíz unitaria.

Date: 09/26/26 Time: 19:50						
Sample (adjusted): 1992 2024						
Included observations: 33 after adjustments						
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	-0.372	-0.372	5.0054	0.025
		2	-0.040	-0.208	5.0660	0.079
		3	0.126	0.037	5.6782	0.128
		4	-0.038	0.029	5.7361	0.220
		5	-0.043	-0.029	5.8123	0.325
		6	-0.217	-0.315	7.8341	0.251
		7	0.195	-0.037	9.5199	0.217
		8	0.085	0.177	9.8539	0.275
		9	-0.197	-0.024	11.717	0.230
		10	0.046	-0.116	11.822	0.297
		11	0.073	-0.059	12.103	0.356
		12	-0.070	-0.058	12.372	0.416
		13	-0.050	-0.002	12.516	0.486
		14	-0.052	-0.111	12.681	0.552
		15	0.226	0.092	15.963	0.385
		16	-0.012	0.182	15.973	0.455

Como tanto en la FAC como en FACP solo el primer rezago es significativo, y el resto no lo son puede tratarse de un ARIMA(1,1,1):

Dependent Variable: D(LN_TOTAL_EXPORT_GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 10/01/26 Time: 19:46				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Failure to improve likelihood (non-zero gradients) after 22 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1992				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.087345	0.029837	2.927374	0.0067
@TREND	-0.002854	0.001577	-1.810139	0.0810
AR(1)	0.248771	0.200656	1.239790	0.2253
MA(1)	-0.942034	0.041625	-22.63118	0.0000
R-squared	0.324623	Mean dependent var		0.023628
Adjusted R-squared	0.252261	S.D. dependent var		0.289719
S.E. of regression	0.250526	Akaike info criterion		0.185958
Sum squared resid	1.757367	Schwarz criterion		0.369175
Log likelihood	1.024672	Hannan-Quinn criter.		0.246689
F-statistic	4.486103	Durbin-Watson stat		1.927918
Prob(F-statistic)	0.010809			
Inverted AR Roots	.25			
Inverted MA Roots	.94			

Saco el componente AR(1) ya que no es significativo:

Dependent Variable: D(LN_TOTAL_EXPORT_GH)
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/01/26 Time: 19:47
Sample (adjusted): 1992 2024
Included observations: 33 after adjustments
Failure to improve likelihood (non-zero gradients) after 23 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1991

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.078934	0.019218	4.107359	0.0003
@TREND	-0.002368	0.001053	-2.247377	0.0321
MA(1)	-0.943477	0.030941	-30.49327	0.0000
R-squared	0.293740	Mean dependent var		0.020227
Adjusted R-squared	0.246655	S.D. dependent var		0.285825
S.E. of regression	0.248083	Akaike info criterion		0.136399
Sum squared resid	1.846351	Schwarz criterion		0.272445
Log likelihood	0.749415	Hannan-Quinn criter.		0.182174
F-statistic	6.238623	Durbin-Watson stat		1.463909
Prob(F-statistic)	0.005426			
Inverted MA Roots	.94			

Date: 10/01/26 Time: 19:48
Sample (adjusted): 1992 2024
Q-statistic probabilities adjusted for 1 ARMA term and 1 dynamic regressor

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 0.214	0.214	1.6477	
		2 0.062	0.017	1.7904	0.181
		3 0.028	0.012	1.8210	0.402
		4 -0.151	-0.168	2.7296	0.435
		5 -0.336	-0.293	7.3967	0.116
		6 -0.363	-0.278	13.023	0.023
		7 -0.023	0.121	13.047	0.042
		8 -0.028	-0.006	13.084	0.070
		9 -0.188	-0.295	14.783	0.063
		10 -0.031	-0.212	14.832	0.096
		11 0.126	-0.012	15.664	0.110
		12 0.043	-0.016	15.768	0.150
		13 0.036	-0.012	15.844	0.198
		14 0.173	-0.027	17.661	0.171
		15 0.189	-0.040	19.952	0.132
		16 -0.048	-0.132	20.109	0.168

*Probabilities may not be valid for this equation specification.

Los tres coeficientes de la estimación son significativos, sin embargo el residuo no es ruido blanco, por lo tanto saco primero la tendencia determinística (y luego el intercepto):

Dependent Variable: D(LN TOTAL EXPORT GH)
 Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
 Date: 09/26/26 Time: 19:55
 Sample (adjusted): 1992 2024
 Included observations: 33 after adjustments
 Failure to improve likelihood (non-zero gradients) after 12 iterations
 Coefficient covariance computed using outer product of gradients
 MA Backcast: 1991

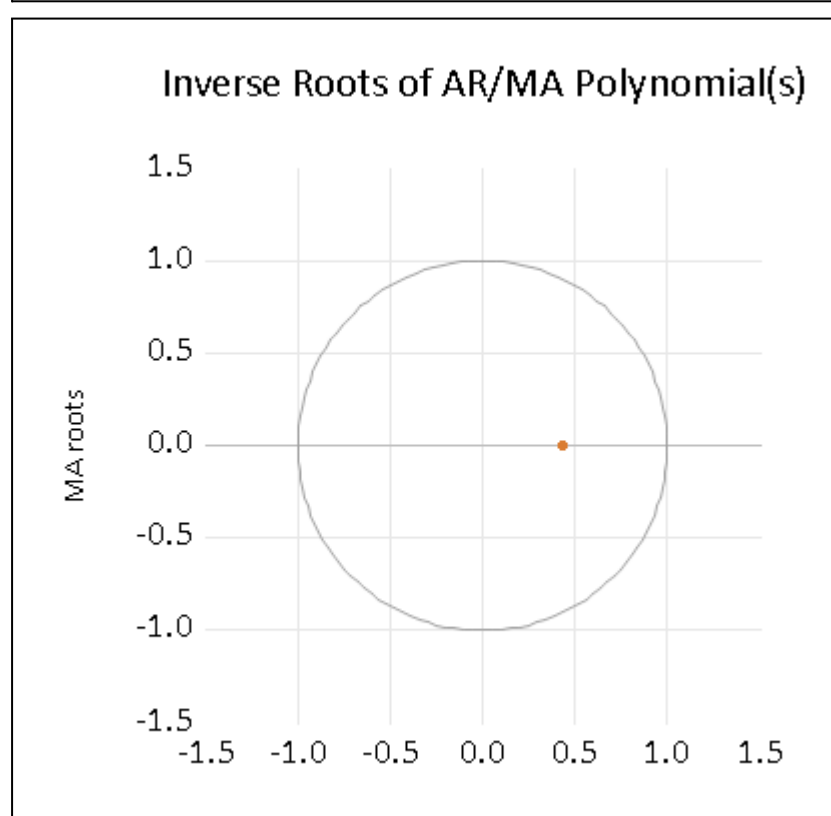
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.028921	0.022108	1.308168	0.2004
MA(1)	-0.540173	0.170880	-3.161116	0.0035
R-squared	0.188953	Mean dependent var		0.020227
Adjusted R-squared	0.162790	S.D. dependent var		0.285825
S.E. of regression	0.261527	Akaike info criterion		0.214135
Sum squared resid	2.120292	Schwarz criterion		0.304833
Log likelihood	-1.533234	Hannan-Quinn criter.		0.244652
F-statistic	7.222182	Durbin-Watson stat		1.842635
Prob(F-statistic)	0.011475			
Inverted MA Roots	.54			

Dependent Variable: D(LN TOTAL EXPORT GH)
 Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
 Date: 09/26/26 Time: 19:56
 Sample (adjusted): 1992 2024
 Included observations: 33 after adjustments
 Failure to improve likelihood (non-zero gradients) after 7 iterations
 Coefficient covariance computed using outer product of gradients
 MA Backcast: 1991

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	-0.433615	0.167110	-2.594789	0.0142
R-squared	0.155625	Mean dependent var		0.020227
Adjusted R-squared	0.155625	S.D. dependent var		0.285825
S.E. of regression	0.262644	Akaike info criterion		0.193800
Sum squared resid	2.207420	Schwarz criterion		0.239149
Log likelihood	-2.197697	Hannan-Quinn criter.		0.209058
Durbin-Watson stat	1.969187			
Inverted MA Roots	.43			

Date: 09/26/26 Time: 19:56
Sample (adjusted): 1992 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.055	-0.055	0.1095	
		2 -0.042	-0.045	0.1757	0.675
		3 0.077	0.072	0.4032	0.817
		4 -0.077	-0.072	0.6415	0.887
		5 -0.169	-0.173	1.8191	0.769
		6 -0.241	-0.285	4.3104	0.506
		7 0.161	0.130	5.4673	0.485
		8 0.083	0.122	5.7892	0.565
		9 -0.173	-0.160	7.2335	0.512
		10 -0.006	-0.156	7.2352	0.613
		11 0.080	-0.014	7.5719	0.671
		12 -0.079	-0.036	7.9195	0.720
		13 -0.071	0.008	8.2079	0.769
		14 0.066	-0.000	8.4746	0.811
		15 0.245	0.159	12.322	0.580
		16 0.017	0.077	12.340	0.653



Ahora si llego a un ARIMA(0,1,1) con ruido blanco y raíz dentro del círculo unitario.

Sin embargo pruebo con el componente AR que antes saque de la estimación debido a que hay veces que AR y MA junto no dan significativos pero si lo son por separado. Luego de que el intercepto y la tendencia determinista no den significativos obtengo:

Dependent Variable: D(LN_TOTAL_EXPORT_GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 09/27/26 Time: 15:10				
Sample (adjusted): 1993 2024				
Included observations: 32 after adjustments				
Convergence achieved after 3 iterations				
Coefficient covariance computed using outer product of gradients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	-0.374019	0.169381	-2.208156	0.0348
R-squared	0.129979	Mean dependent var		0.023628
Adjusted R-squared	0.129979	S.D. dependent var		0.289719
S.E. of regression	0.270235	Akaike info criterion		0.251704
Sum squared resid	2.263841	Schwarz criterion		0.297508
Log likelihood	-3.027263	Hannan-Quinn criter.		0.266887
Durbin-Watson stat	2.089343			
Inverted AR Roots	-0.37			

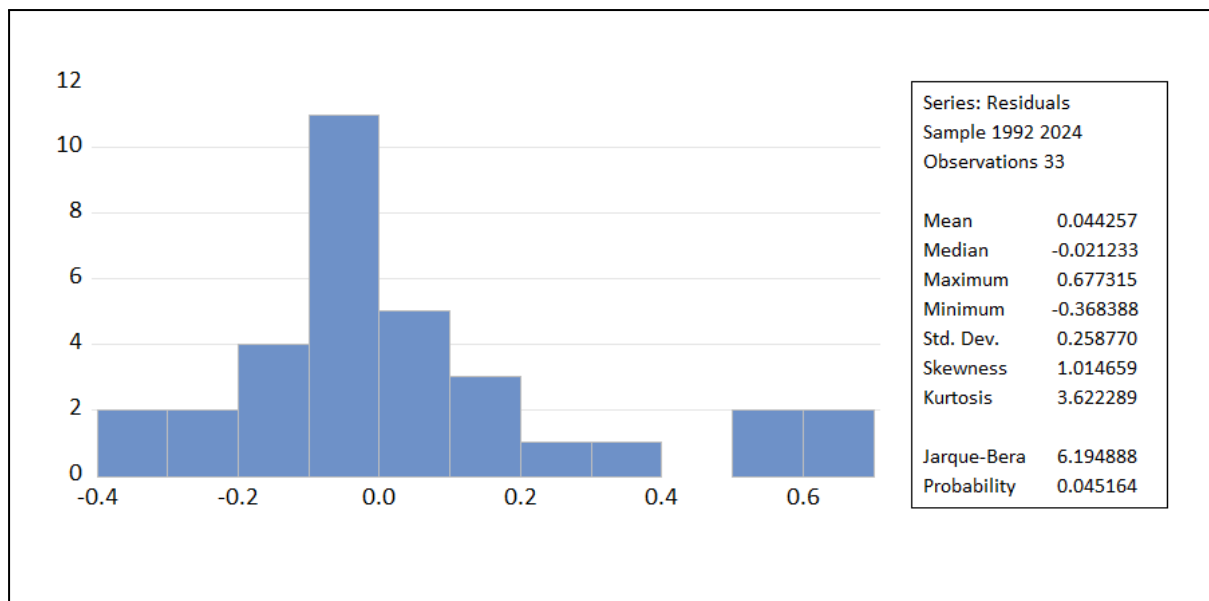
Date: 09/27/26 Time: 15:11						
Sample (adjusted): 1993 2024						
Q-statistic probabilities adjusted for 1 ARMA term						
Autocorrelation		Partial Correlation		AC	PAC	Q-Stat Prob
		1	-0.093	-0.093	0.3050	
		2	-0.177	-0.188	1.4452	0.229
		3	0.122	0.089	2.0071	0.367
		4	0.011	-0.001	2.0116	0.570
		5	-0.198	-0.168	3.5845	0.465
		6	-0.236	-0.302	5.9151	0.315
		7	0.215	0.109	7.9192	0.244
		8	0.123	0.134	8.6057	0.282
		9	-0.216	-0.117	10.809	0.213
		10	-0.002	-0.113	10.809	0.289
		11	0.117	-0.048	11.524	0.318
		12	-0.067	-0.026	11.766	0.381
		13	-0.116	0.004	12.540	0.403
		14	0.048	-0.023	12.679	0.473
		15	0.251	0.145	16.726	0.271
		16	-0.020	0.076	16.754	0.334

Esta estimacion tambien arroja coeficientes significativos, ruido blanco, y la raiz dentro del circulo unitario, por lo tanto utilizo el criterio de Schwarz para comparar entre los modelos ARIMA(1,1,0) y ARIMA(0,1,1), sin embargo para hacerlo debo asegurarme de que ambas estimaciones tengan la misma cantidad de observaciones, por lo tanto estimo nuevamente el ARIMA(0,1,1) con 32 observaciones al igual que el ARIMA(1,1,0).

Dependent Variable: D(LN_TOTAL_EXPORT_GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 10/01/26 Time: 19:56				
Sample: 1993 2024				
Included observations: 32				
Failure to improve likelihood (non-zero gradients) after 6 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1992				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	-0.435560	0.168873	-2.579217	0.0149
R-squared	0.153200	Mean dependent var		0.023628
Adjusted R-squared	0.153200	S.D. dependent var		0.289719
S.E. of regression	0.266605	Akaike info criterion		0.224651
Sum squared resid	2.203419	Schwarz criterion		0.270456
Log likelihood	-2.594420	Hannan-Quinn criter.		0.239834
Durbin-Watson stat	1.954209			
Inverted MA Roots	.44			

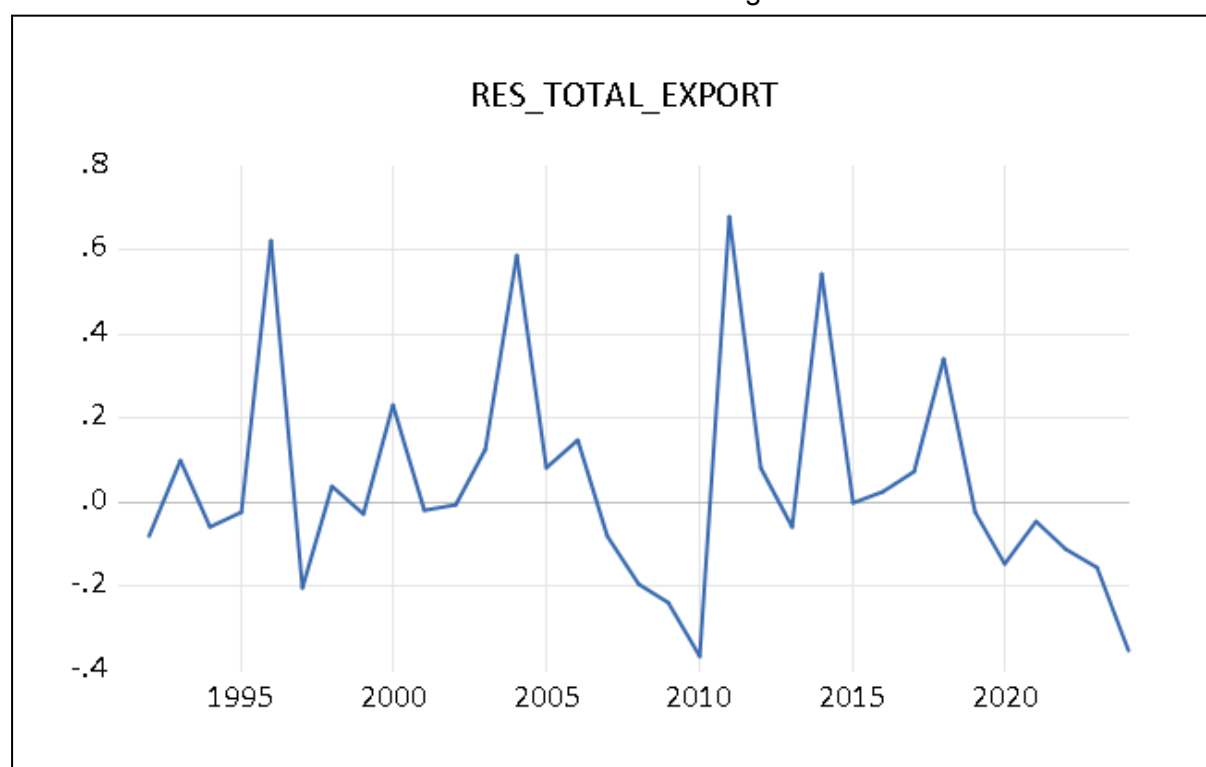
Date: 10/01/26 Time: 19:57					
Sample: 1993 2024					
Q-statistic probabilities adjusted for 1 ARMA term					
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.052	-0.052	0.0939	
		2 -0.049	-0.052	0.1815	0.670
		3 0.074	0.069	0.3842	0.825
		4 -0.046	-0.042	0.4674	0.926
		5 -0.188	-0.187	1.8888	0.756
		6 -0.246	-0.288	4.4215	0.490
		7 0.157	0.118	5.4914	0.483
		8 0.093	0.136	5.8869	0.553
		9 -0.179	-0.158	7.4075	0.493
		10 -0.009	-0.148	7.4117	0.594
		11 0.088	-0.028	7.8129	0.647
		12 -0.047	-0.004	7.9329	0.719
		13 -0.068	0.034	8.1998	0.769
		14 0.074	0.021	8.5312	0.807
		15 0.238	0.136	12.163	0.593
		16 0.001	0.057	12.163	0.667

Con la nueva estimación del ARIMA(0,1,1) obtengo un Schwarz de 0.2704, y con el ARIMA(1,1,0) un Schwarz de 0.2975, por lo tanto el modelo elegido es el ARIMA(0,0,1) sin intercepto ni tendencia.



Con el test de Jarque-Bera observamos que los residuos del ARIMA(0,1,1) no se comportan normalmente con un nivel de significancia del 5%.

Utilizo los residuos de ARIMA(0,1,1) y creo una nueva serie a partir de estos, que como son ruido blanco va a ser estacionario en torno a una media igual a 0.



Date: 10/01/26 Time: 20:06

Sample (adjusted): 1992 2024

Included observations: 33 after adjustments

Included observations: 55 after adjustments		AC	PAC	Q-Stat	Prob	
Autocorrelation	Partial Correlation					
		1	-0.055	-0.055	0.1095	0.741
		2	-0.042	-0.045	0.1757	0.916
		3	0.077	0.072	0.4032	0.940
		4	-0.077	-0.072	0.6415	0.958
		5	-0.169	-0.173	1.8191	0.874
		6	-0.241	-0.285	4.3104	0.635
		7	0.161	0.130	5.4673	0.603
		8	0.083	0.122	5.7892	0.671
		9	-0.173	-0.160	7.2335	0.613
		10	-0.006	-0.156	7.2352	0.703
		11	0.080	-0.014	7.5719	0.751
		12	-0.079	-0.036	7.9195	0.791
		13	-0.071	0.008	8.2079	0.830
		14	0.066	-0.000	8.4746	0.863
		15	0.245	0.159	12.322	0.655
		16	0.017	0.077	12.340	0.720